# Extreme weather risk knowledge organiser

Tuesday 16 March 2027 | P2 | 40 minutes | Year A

Humanities | LAUNCH | Spring 2 Week 4 | HUM-LAUNCH-S2-W04

White paper version. Read quietly, point, write, type, speak privately or direct a scribe who reads your words back. You may pass. No public answer or reading aloud is required.

Learning outcome: Extreme-weather hazards & risk.

Extreme weather can include heavy rain, strong wind or extreme heat. This lesson focuses on heavy rain and flooding. Risk is possible harm. It depends on the hazard and who or what is exposed, vulnerability and capacity. Consider likelihood and potential severity together.

| Idea | Meaning in this lesson |
|---|---|
| Hazard | A process that can cause harm: heavy rain and resulting flooding. |
| Exposure | People, buildings, routes or services in the affected area. |
| Vulnerability | Conditions that increase harm, such as low equipment or limited drainage. |
| Capacity | Strengths and resources that help people or systems cope. |
| Likelihood | How likely flooding is under the given assumptions. |
| Severity | How serious the impacts could be if flooding occurs. |

## Invented comparison

A: high likelihood, major potential severity. B: high likelihood, minor severity. C: medium likelihood, major severity. These are given teaching assumptions, not measured probabilities or an actual forecast.

## Different risk from the same rain

A has homes and a water pump exposed, older drains and low equipment. B has planned water-storage space and an upper path. Capacity and exposure differ. C relies on one access bridge.

## A measure and its limit

Improved drains can help with surface water but cannot stop every river flood. Protected equipment can reduce one service risk while other risks remain. Measures need resources and maintenance.

Where this leads: Meteorologists, flood planners and engineers use evidence about weather, places and services to explain risk and compare plans.